

MOHAMED SHAMIL

AI/ML Engineer

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SUMMARY

AI/ML Engineer focused on building practical AI applications using LLMs, RAG, and machine learning. Built and deployed AI agents, backend services, and RAG systems during a 6-month AI/ML internship at CodeGen International. Experienced in building AI systems with tool use, data retrieval, testing, and reliability checks. Currently researching hybrid LLM-based reasoning for multi-step problem solving.

EXPERIENCE

CodeGen International (Pvt) Ltd

Mar 2025 – Sep 2025

AI/ML Engineer Intern

Colombo, Sri Lanka

- Shipped production **MCP and RAG** agent systems into live client workflows, replacing processes previously handled by hand: agents retrieve domain knowledge, select tools, and carry a task to completion rather than returning a single answer.
- Built LLM agents with **LangChain** - persistent memory, tool selection, and autonomous execution - so a task could span multiple turns and tool calls without losing state, backed by **FastAPI** and **PostgreSQL** services holding agent state and conversation history across live sessions.
- Reduced repeated failure modes on domain-specific tasks through systematic prompt engineering and fine-tuning, iterating against real production conversations rather than benchmark prompts.
- Containerized with **Docker** and deployed to **AWS**, integrating the **WhatsApp Business API** so end users reached the agents through a channel they already used.
- Tools & Technologies:** Python, FastAPI, LangChain, MCP, RAG, LLMs, WhatsApp Business API, PostgreSQL, Docker, AWS.

RESEARCH

Hybrid Architecture of LLM and HRM for Multi-Step Mathematical Problem Solving

2025 – Present

Final-Year Research Project – University of Moratuwa

Advisor: Dr. Romesh Thanuja

- Designed a neuro-symbolic hybrid reasoning architecture combining LLM-based semantic parsing, graph-structured intermediate representations, and a Hierarchical Reasoning Model (HRM) for interpretable multi-step mathematical problem solving.
- Developed a dependency-graph bridge encoder that transforms natural-language math word problems into executable reasoning traces with auditable intermediate computations.
- Implemented HRM high, low-level transformer modules, digit-level arithmetic decoders, Adaptive Computation Time (ACT) halting, and curriculum learning pipelines in PyTorch.
- Trained and evaluated the system on GSM8K (7,473 train / 1,319 test), achieving **88.5% exact-match accuracy** (89.7% on GSM8K-Platinum).
- Achieved 99% executor correctness and 100% explanation faithfulness, demonstrating reliable and interpretable multi-step reasoning.
- Stack:** PyTorch, Transformers, HRM, OpenAI API (GPT-4o Mini), FastAPI, React

PROJECTS

Enterprise AI Support Platform | Python, FastAPI, Qdrant, n8n, PostgreSQL, Docker | GitHub

2026

- Built a **policy-gated support agent** that separates LLM planning from authorization and execution, ensuring **prompt injection cannot bypass deterministic Python policy checks**, EXECUTE-tier actions such as refunds are blocked before reaching tool handlers.
- Built a **bounded ReAct**, **RAG pipeline** over 15 schema-validated tools using hybrid dense/BM25 retrieval, RRF, reranking, parent-context expansion, claim verification, and human escalation, achieved **Recall@1 0.834**, **MRR 0.931**, and **nDCG@3 0.949** on 350 benchmark cases.
- Added **risk-based routing** that combines policy outcomes, retrieval quality, groundedness, action risk, and sentiment to route tickets to **AUTOMATE**, **ASSIST**, or **ESCALATE**, with hard safety gates for high-risk or unevaluable cases.
- Built **reliability and evaluation infrastructure** with failure-safe retrieval fallbacks, claim-level groundedness checks, reproducible benchmarking, audit logging, and token/latency tracking, deterministic intent classification outperformed the LLM (**0.889 vs 0.837 accuracy**), while unsupported claims were routed to human review.

- Built a **policy-gated hospital AI agent** that separates LLM planning from authorization and execution; tested **19 adversarial scenarios**, closing **15 unauthorized-execution paths** that previously bypassed security controls.
- Implemented **LangGraph** , **MCP workflows** with human approval, resumable execution, strict schemas, bounded retries, and idempotency, including automated patient-admission and operational workflows.
- Built **hospital workflow automation and document generation**, including meeting/appointment automation and validated PDF generation to verify patient and appointment details before producing reports.
- Designed and developed the **end-to-end system architecture** across React, FastAPI, PostgreSQL, Docker, and AWS, with **342 automated tests** covering backend, frontend, security, integration, and end-to-end workflows.
- Built a **PostgreSQL audit trail** covering authentication, policy checks, approvals, execution, and failures, with argument-bound approvals to prevent replay and database locking for safe concurrent updates.

- Built a **WhatsApp AI sales agent** with tool-based RAG over a **200-SKU product catalog**, SQLite conversation memory, HMAC webhook authentication, and latency/cost tracking.
- Evaluated **four retrieval strategies** on 100 labelled queries and implemented **query-adaptive retrieval**, using hybrid RRF for specification queries (**0.92 hit@3**) and dense retrieval for semantic queries (**0.68 hit@3**).
- Added **out-of-catalog detection and abstention** to prevent unsupported product answers, and improved reliability by fixing a concurrency bug that lost **190/200 messages** and expanding the test suite from **36 to 118 passing tests**.

- Built a **weapon detection system using YOLOv8** that combines object detection with scene context to estimate threat levels.
- Replaced a **crisp threshold-based scorer with a Mamdani fuzzy-logic system**, combining weapon confidence, crowd density, scene sensitivity, and lighting into a continuous threat score.
- Improved weapon-scene detection from **13/18 to 18/18** and reduced simulated false alarms from **24.4% to 2.2%**, with leakage-controlled evaluation and **46 regression tests**.

- Developed, in a **Sysco Labs**-mentored team, ML system analyzing CO2e emissions across the food supply chain to surface sustainability insights, training Scikit-learn models on cleaned, feature-engineered emissions data.
- Built backend microservices for vendor and product management and interactive data-visualization components for the emissions dashboard.

EDUCATION

University of Moratuwa	2022 – 2026
BSc (Hons) in Artificial Intelligence	Colombo, Sri Lanka
• Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Science, and Algorithms.	
Kekunagolla National School	2018 – 2020
G.C.E. Advanced Level (Physical Science) – Mathematics A, Chemistry A, Physics B	Kurunegala, Sri Lanka

TECHNICAL SKILLS

Agentic AI & LLMs:	MCP, LangGraph, LangChain, RAG, n8n, Microsoft Foundry, multi-agent orchestration, tool authorization, evaluation harnesses, prompt engineering, fine-tuning, LLM APIs
ML & Deep Learning:	PyTorch, Transformers, Hugging Face, sentence-transformers, Scikit-learn, XGBoost, LightGBM, Computer Vision (YOLOv8, OpenCV)
Programming Languages:	Python, Java, C, JavaScript
Backend & Data:	FastAPI, Flask, Spring Boot, RESTful APIs, SQLAlchemy, PostgreSQL, MySQL, Qdrant
Frontend & AI Apps:	React, Flutter, Streamlit
Cloud & Infrastructure:	Docker, AWS, GCP, CI/CD (GitHub Actions), Git

ADDITIONAL

Languages:	English (Fluent) · Tamil (Native) · Sinhala (Fluent)
Involvement:	IEEE Student Member · Company Coordinator, Career Fair 2024, University of Moratuwa · MoraXtreme 8.0 & CodeRush 2023 (inter-university competitive programming)